Practicum No.10 **Date:……….**

Temperature Display System

I. Practical Significance

Temperature control is a fundamental requirement in industrial automation, HVAC systems, chemical processing, biomedical equipment, and smart manufacturing. Precise closed-loop temperature regulation ensures product quality, process safety, and energy efficiency. This practicum introduces students to designing and simulating a Temperature Control System in MATLAB Simulink using the Simscape Thermal library combined with a continuous-time PID Controller.

The model mirrors a real-world closed-loop control scenario: a Setpoint (Constant block) defines the desired temperature, the PID Controller computes the corrective action, a Controlled Temperature Source applies the heat input, and a Temperature Sensor measures the actual output. The error signal continuously drives the controller to minimize deviation from the setpoint. By simulating this system, students develop the critical skill of translating physical control theory into executable, verifiable models — directly applicable to industrial process control, embedded thermostat design, and PLC programming.

II. Competency and Industry-Specific Practical Skills

- This practicum builds the following control systems and simulation skills:
- Use the principles of feedback control to design stable and responsive temperature regulation systems.
- Model closed-loop control systems in MATLAB Simulink using Simscape Thermal blocks (PDO).
- Configure and tune a PID controller (Kp, Ki, Kd) for temperature regulation (PDO).
- Interpret Scope output to evaluate system response — rise time, overshoot, and steady-state error (CDO).
- Connect Simulink signal-domain blocks with Simscape physical-domain blocks using PS converters (PDO).
- Follow safe simulation practices and maintain disciplined model documentation (ADO).

III. Relevant CO(s) *(As stated in the Course Structure)*

- **CO3**: Apply control system principles to design, simulate, and analyse feedback control loops using MATLAB Simulink and Simscape toolboxes.

IV. Practical Outcome (PrO) *(As stated in the Course Structure)*

- **PrO**: Design and simulate a closed-loop Temperature Control System in Simulink using a PID Controller and Simscape Thermal blocks; analyse the transient response from the Scope output.

V. Related ADO(s) *(As stated in the Course Structure)*

- Follow safe laboratory and software-use practices.
- Handle MATLAB/Simulink licenses and model files responsibly.
- Work collaboratively to debug simulation errors.
- Maintain clean block diagrams with meaningful labels.

- Follow ethical practices in reporting simulation observations.

VI. Minimum Underpinning Theory *(Include relevant content & images for this practicum.)*

1. Closed-Loop (Feedback) Control

In a closed-loop system the output is continuously measured and fed back to the input. The controller uses the error (difference between setpoint and measured output) to generate a corrective action. This self-correcting mechanism ensures the system reaches and maintains the desired setpoint despite disturbances.

2. PID Controller

The PID (Proportional–Integral–Derivative) controller computes the control signal u(t) as:

$u(t) = K_p \cdot e(t) + K_i \cdot \int e(t)dt + K_d \cdot (de/dt)$

In this model the PID is configured in parallel form with Kp = 0.9, Ki = 0.4, and Kd = 0 (PI controller), operating in continuous time using the Forward Euler integration method.

Term	Parameter	Effect on Response
Kp = 0.9	Proportional gain	Speeds up response; may cause steady-state error
Ki = 0.4	Integral gain	Eliminates steady-state error
Kd = 0	Derivative gain (disabled)	Not used; reduces overshoot when non-zero

3. Simscape Thermal Domain Blocks

Simscape models physical systems using conserved quantities (heat flow) through connection lines. Key blocks in this model:

Block	Description
Controlled Temperature Source	Generates a thermal input whose temperature is set by the PID output (via Simulink-PS Converter). Ports: A and B (thermal), control input.
Temperature Sensor	Measures the temperature difference across its thermal ports (A, B) and outputs a Physical Signal (T port).
Thermal Reference	Sets the thermal ground (0 K reference) for the Simscape network.
Solver Configuration	Provides numerical solver settings for the Simscape network (RelTol = 0.001).

4. Signal-Domain to Physical-Domain Conversion

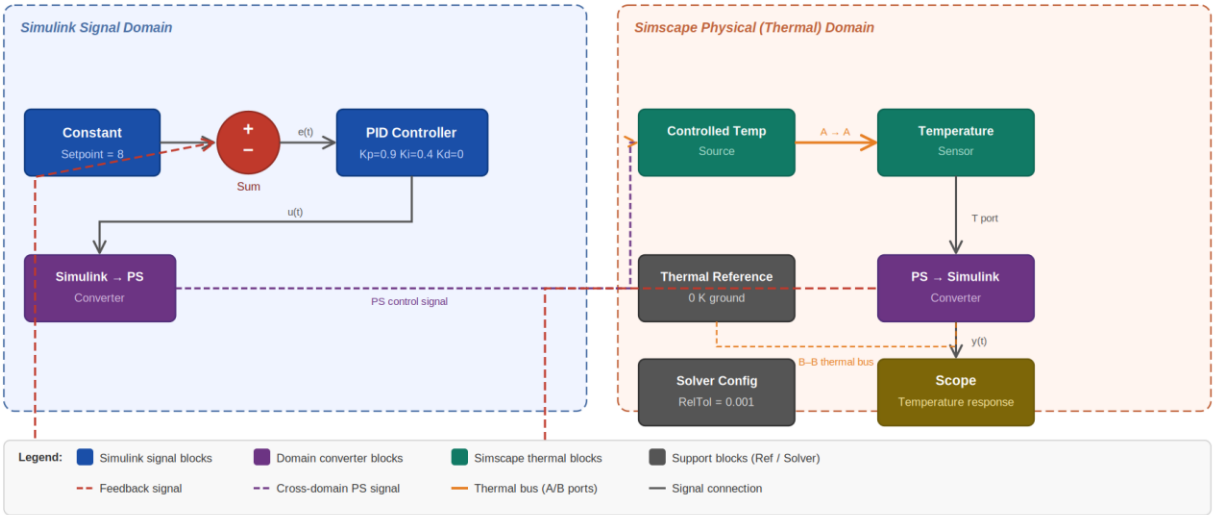
Simulink operates on dimensionless signals whereas Simscape uses typed physical signals. Two converter blocks bridge the domains:

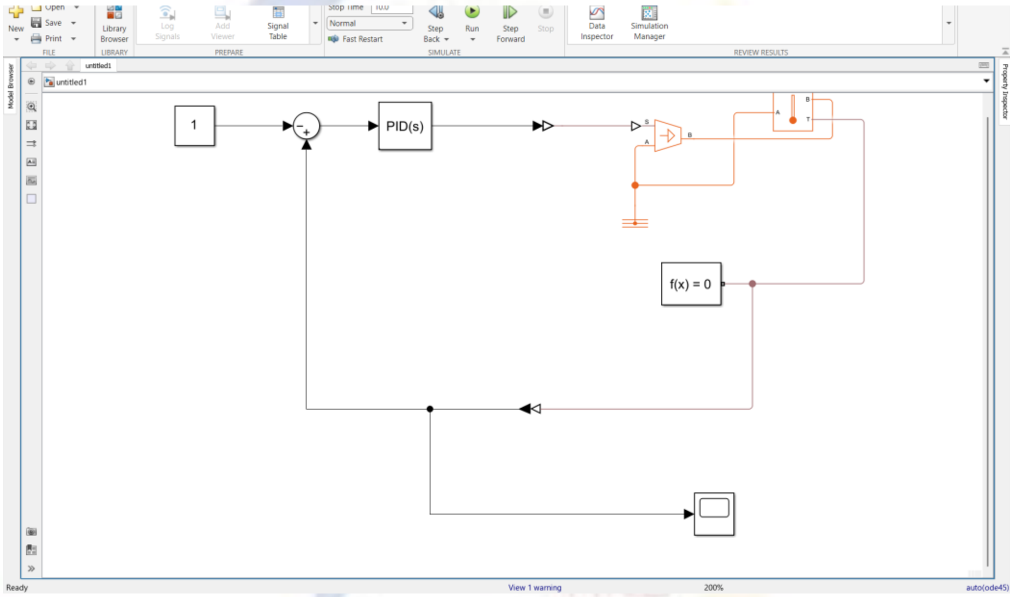
- Simulink-PS Converter: converts the PID output (Simulink signal) into a Physical Signal for the Controlled Temperature Source.
- PS-Simulink Converter: converts the Temperature Sensor Physical Signal output back into a Simulink signal for the Sum block (feedback) and the Scope.

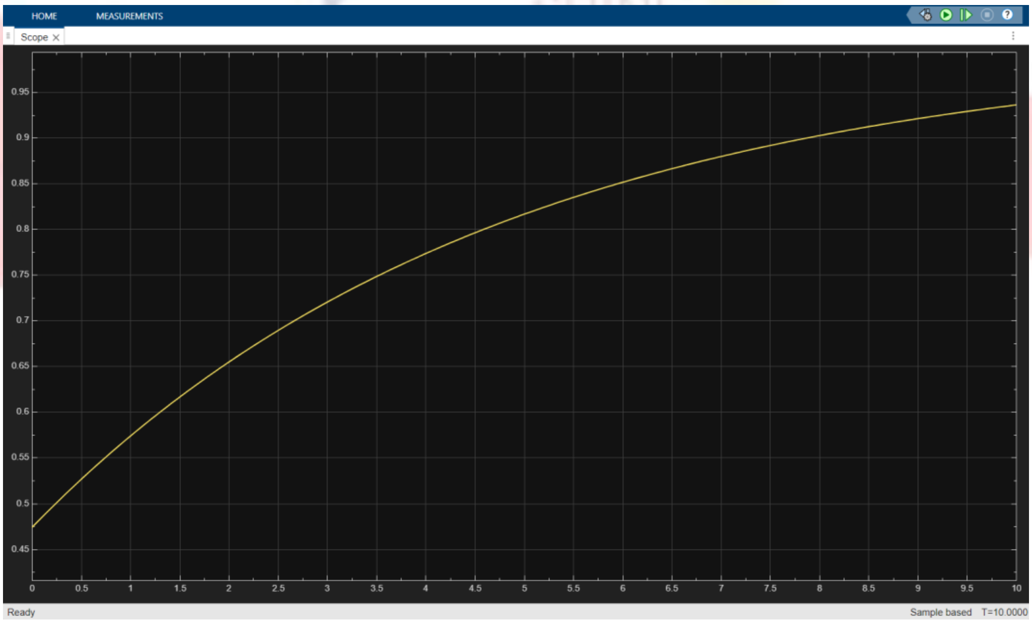
5. Signal Flow in the Model

Step	Source Block	Destination Block	Signal / Connection
1	Constant (Value = 8)	Sum (+)	Setpoint temperature reference
2	PS-Simulink Converter	Sum (−)	Measured temperature fed back
3	Sum	PID Controller	Error signal e(t) = setpoint − measured
4	PID Controller	Simulink-PS Converter	Control action u(t)
5	Simulink-PS Converter	Controlled Temperature Source (S port)	Physical control signal
6	Temperature Sensor (T port)	PS-Simulink Converter	Measured temperature (Physical Signal)
7	PS-Simulink Converter	Scope	Temperature response for display

VII. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart *(Include probable photos of experimental set-up.)*







VIII. Resources Required

S.No.	Resource	Suggested Specification	Qty
1	MATLAB & Simulink	R2021b or later	1
2	Simscape Toolbox	Part of MATLAB Simscape add-on	1
3	Simscape Foundation Library (fl_lib)	Thermal domain blocks	1
4	Simulink PID Controller block (slpidlib)	PID, 1DOF, continuous	1
5	TemperatureControlSystem.slx	Provided model file	1
6	Computer / Workstation	Windows 10/11 or Linux, 8 GB RAM	1
7	MATLAB Documentation	Simscape Thermal & PID Tuner guides	1

IX. Safety Precautions

- Ensure MATLAB and Simscape licenses are valid before starting; unlicensed toolboxes cause simulation errors.
- Save the model file before running simulation to prevent loss of changes.
- Do not modify PID gain values or solver tolerances without understanding their effect on stability.
- Keep simulation stop time within reasonable bounds (e.g., 100 s) to avoid excessive computation time.
- Do not alter the Simscape Solver Configuration block parameters unless directed.

X. Procedure *(Self-Instructional)*

1. Launch MATLAB and verify that Simulink and Simscape toolboxes are available.
2. Navigate to the folder containing TemperatureControlSystem.slx using the MATLAB Current Folder panel.
3. Type open('TemperatureControlSystem.slx') in the Command Window or double-click the file to open it in Simulink.
4. Study the block diagram. Identify the following blocks and their interconnections: Constant (setpoint), Sum (error calculation), PID Controller, Simulink-PS Converter, Controlled Temperature Source, Temperature Sensor, PS-Simulink Converter, Scope, Thermal Reference, and Solver Configuration.
5. Double-click each block to examine its parameters. Record the PID gains (Kp, Ki, Kd) and the Constant value in your observation sheet.
6. Trace the signal path: Constant → Sum (+) → PID → Simulink-PS Converter → Controlled Temperature Source → Temperature Sensor → PS-Simulink Converter → Sum (−) [feedback] and Scope.
7. Identify the error signal at the Sum block output. Confirm the Sum block input signs are |−+ (first input is positive setpoint, second is negative feedback).
8. Note that the PS-Simulink Converter and Simulink-PS Converter blocks serve as bridges between the Simulink signal domain and the Simscape physical (thermal) domain.

9. Confirm the Thermal Reference block provides the ground node for the Simscape thermal network.
10. Verify that the Solver Configuration block is connected to the thermal network.
11. Open Simulation > Model Configuration Parameters (Ctrl+E).
12. Set the Stop Time to 100 seconds.
13. Select a variable-step solver (e.g., ode23t or ode45) suitable for stiff systems with Simscape blocks.
14. Set the Relative Tolerance to 1e-3 and click OK.
15. Ensure the Scope block is visible (double-click to open the Scope window before running).
16. Click the Run button (green triangle) or press Ctrl+T to start the simulation.
17. Observe the Scope window. The waveform represents the measured temperature over time.
18. Allow the simulation to complete. The system should reach a steady state close to the setpoint value of 8.
19. Use the Scope toolbar to zoom in on the transient region (0–20 s) to clearly observe the rise time and any overshoot.
20. Note the final steady-state temperature value from the Scope. Record the approximate Y-axis range (expected: approximately 3.3 to 7.9 based on model data).
21. Open the PID Controller block (double-click) and record Kp=0.9, Ki=0.4, and confirm Kd=0 (PI controller).
22. Note the effect of the integral term (Ki) — it eliminates steady-state error, driving the output to the setpoint.
23. Change Kp to 1.5 and re-run the simulation. Observe any change in rise time or overshoot. Record the result.
24. Restore Kp to 0.9. Now change Ki to 0.1 and re-run. Observe the slower error elimination. Record the result.
25. Restore all parameters to original values (Kp=0.9, Ki=0.4, Kd=0) before proceeding Double-click the Constant block and change its value from 8 to 15.
26. Re-run the simulation and observe the new steady-state temperature on the Scope.
27. Compare rise time and settling behaviour with the previous setpoint.
28. Restore the Constant value to 8 after recording observations.
29. Take a screenshot of the Scope output showing the full temperature response.
30. Save the Simulink model (Ctrl+S).
31. Record all observations, Scope screenshots, and parameter values in the journal.
32. Prepare a labelled block diagram sketch of the model for inclusion in the report.
33. Finalise the report with code/model parameters, screenshots, and your analysis of the control response.

XI. Actual Resources Used

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
1.	Laptop	Asus	Intel i5, 8GB RAM, Windows 11	1	Used for simulation
2.	MATLAB Software	MathWorks	MATLAB with Simulink	1	Installed in lab
3.	Simscape Toolbox	MathWorks	Thermal modelling library	1	Used for physical blocks
4.	Simulink PID Controller	MathWorks	Continuous PID controller	1	Used for control system

XII. Procedure Followed

1. MATLAB software was opened and Simulink environment was launched.
2. A new Simulink model was created for the temperature display system.
3. Required blocks such as Constant, Sum, PID Controller, Simulink-PS Converter, Controlled Temperature Source, Temperature Sensor, PS-Simulink Converter, Scope, Thermal Reference and Solver Configuration were added.
4. The blocks were connected to form a closed-loop temperature control system.
5. The Constant block was set to a desired temperature value of 8.
6. PID controller parameters were configured to regulate the temperature.
7. Simulation settings were adjusted and the model was executed.
8. The temperature response was observed using the Scope block.

XIII. Results/Observations

1. The temperature display system successfully monitored and regulated the temperature using a feedback control loop.
2. The temperature gradually increased and stabilized near the setpoint value, demonstrating effective PID control.

XIV. Interpretation of Results

The simulation demonstrates how a feedback control system regulates temperature automatically. The PID controller continuously adjusts the control signal based on the error between desired and measured temperature, ensuring stable and accurate temperature control.

XV. Conclusions

The temperature display system using Simulink and Simscape was successfully modeled and simulated. The PID controller effectively maintained the temperature at the desired setpoint, demonstrating the importance of feedback control in temperature regulation systems.

XVI. Suggested Practicum Related Questions:

***Note:** Below are a few questions related to industry-specific higher-order thinking skills (HOT) that teachers must use to ensure students achieve the predetermined course outcomes.*

1. Compare P, PI, and PID controllers for temperature regulation: analyse the trade-offs between rise time, steady-state error, and overshoot using the Simulink Scope output.
2. Evaluate the role of the Simulink-PS and PS-Simulink Converter blocks: justify why direct connections between Simulink signal lines and Simscape physical ports are not permitted.
3. Identify the effect of increasing the derivative gain (Kd) on the temperature response in this model, and explain why Kd is set to zero in the current configuration.
4. Correlate the Solver Configuration block parameters (RelTol, AbsTol, MinStep) with the numerical accuracy and simulation speed of the Simscape thermal network.
5. Propose how this closed-loop temperature control model could be extended to include a thermal mass (heat capacity) block and a disturbance input, and predict the impact on the PID response.

XVIII References / Suggestions for further reading

1. MATLAB Simulink Documentation, The MathWorks Inc., latest edition.
2. Simscape User's Guide – Thermal Domain, The MathWorks Inc.

3. Åström, K. J. and Hägglund, T., *PID Controllers: Theory, Design and Tuning*, Instrument Society of America, 2nd edition.
4. Ogata, K., *Modern Control Engineering*, Pearson Education, 5th edition.
5. PID Tuner App Documentation, The MathWorks Inc., MATLAB Control System Toolbox.
6. Franklin, G. F., Powell, J. D., and Emami-Naeini, A., *Feedback Control of Dynamic Systems*, Pearson, 8th edition.
7. Online Tutorial – PID Temperature Control using MATLAB Simulink and Simscape (YouTube/MathWorks MATLAB Tech Talks).

XIX Suggested Assessment Scheme

The performance indicators given serve as a guideline for assessing the *'process-related skills/LOs'* (marks to be awarded in real-time in the laboratory by the faculty member, as these cannot be measured after the practicum is over) and *'product-related skills/LOs'*

Note: The weightage for the ***process-related* skills** and ***product-related* skills** will change depending on the PrO, COs, and the competency related to the concerned course.

	Performance Indicators	Maximum Marks	Marks Obtained
Process-related Skills: 18 Marks - 60%			
1	Handling of samples / Mechanical Tools / Microscope properly.	6	
2	Choice of relevant measurements.	5	
3	Follow safety precautions/protocols.	4	
4	Contribution as a team member (Rubric to be developed).	3	
Product-related Skills: 12 Marks - 40%			
1	Follow the principles underlying sample preparation.	3	
2	Follow the principles of the Interpretation of the images.	2	
3	Interpretation of Results.	3	
4	Conclusions.	2	
5	Practical related questions.	1	
6	Submitting the Journal on time.	1	
	Total	**30**	

To be filled by Student:

S.No.	Name of the Student	Register No.
1	**Neavis Rishva. J**	**URK25CS1236**

To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Date of Evaluation
Process-Related Skills (18)	Product-Related Skills (12)	Total (30)		

****